

MAT61-4

Stochastic Calculus

Brownian Motion, Itô's Formula, SDEs, and the Black-Scholes
Model

The mathematical language of continuous-time uncertainty, from Brownian motion to
the pricing of derivatives.

Albert School — 30 hours

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1 Stochastic processes

A **stochastic process** is a collection of random variables $\{X_t\}$ indexed by time t , either discrete ($t \in \{0, 1, 2, \dots\}$) or continuous ($t \geq 0$).

Definition 1 (Random walk) Let $\xi_1, \xi_2, \dots$ be independent with $P(\xi_i = +1) = p$ and $P(\xi_i = -1) = 1 - p$. The **random walk** is $S_0 = 0$, $S_n = \sum_{i=1}^n \xi_i$. It is **symmetric** if $p = 1/2$ and **biased** otherwise. Then $\mathbb{E}[S_n] = n(2p - 1)$ and, when $p = 1/2$, $\text{Var}(S_n) = n$.

Definition 2 (Markov chain) A process $\{X_n\}$ on a finite or countable state space $\mathcal{S}$ is a **Markov chain** if

$$P(X_{n+1} = j \mid X_n = i, X_{n-1}, \dots, X_0) = P(X_{n+1} = j \mid X_n = i) =: P_{ij}.$$

The **transition matrix** is $P = (P_{ij})$, with $P_{ij} \geq 0$ and $\sum_j P_{ij} = 1$ for every row i .

Definition 3 (Stationary distribution) A row vector $\pi = (\pi_i)$ with $\pi_i \geq 0$ and $\sum_i \pi_i = 1$ is a **stationary distribution** of the chain if

$$\pi P = \pi, \quad \text{i.e.} \quad \pi_j = \sum_i \pi_i P_{ij} \text{ for all } j.$$

If the chain is distributed as π at one step, it is distributed as π at every later step.

Definition 4 (Conditional expectation) For a random variable X and a sub- σ -algebra (information set) $\mathcal{G}$, the **conditional expectation** $\mathbb{E}[X \mid \mathcal{G}]$ is the $\mathcal{G}$ -measurable random variable satisfying

$$\mathbb{E}[\mathbb{E}[X \mid \mathcal{G}] \mathbf{1}_A] = \mathbb{E}[X \mathbf{1}_A] \quad \text{for all } A \in \mathcal{G}.$$

It satisfies the **tower property** $\mathbb{E}[\mathbb{E}[X \mid \mathcal{G}]] = \mathbb{E}[X]$ and is linear.

Definition 5 (Martingale) A process $\{M_t\}$ adapted to a filtration $\{\mathcal{F}_t\}$, with $\mathbb{E}[|M_t|] < \infty$, is a **martingale** if

$$\mathbb{E}[M_t \mid \mathcal{F}_s] = M_s \quad \text{for all } s \leq t.$$

The conditional expectation of a future value, given present information, equals the present value (the “fair game” property).

2 Brownian motion

Definition 6 (Brownian motion (four properties)) A process $\{W_t\}_{t \geq 0}$ is a **standard Brownian motion** (Wiener process) if:

1. $W_0 = 0$;
2. it has **independent increments**: for $t_0 < t_1 < \dots < t_n$, the increments $W_{t_1} - W_{t_0}, \dots, W_{t_n} - W_{t_{n-1}}$ are independent;
3. it has **Gaussian increments**: $W_t - W_s \sim \mathcal{N}(0, t - s)$ for $s < t$;
4. it has **continuous paths**: $t \mapsto W_t$ is continuous almost surely.

Proposition 7 (Covariance function) For a standard Brownian motion,

$$\text{Cov}(W_s, W_t) = \min(s, t).$$

Proof. Take $s \leq t$. Then $W_t = W_s + (W_t - W_s)$, with $W_t - W_s$ independent of W_s and $\mathbb{E}[W_s] = 0$. Hence $\text{Cov}(W_s, W_t) = \mathbb{E}[W_s W_t] = \mathbb{E}[W_s^2] + \mathbb{E}[W_s(W_t - W_s)] = \mathbb{E}[W_s^2] = s = \min(s, t)$. $\square$

Proposition 8 (Quadratic variation) On $[0, t]$, partition into n equal subintervals of width $\Delta t = t/n$. Then

$$\sum_{k=1}^n (W_{t_k} - W_{t_{k-1}})^2 \rightarrow t \quad \text{as } n \rightarrow \infty,$$

in the mean-square sense. Symbolically, $(dW)^2 = dt$.

Proposition 9 (Non-differentiability) The paths of Brownian motion are almost surely continuous but nowhere differentiable. The increment $W_{t+\Delta t} - W_t$ has standard deviation $\sqrt{\Delta t}$, so the difference quotient $(W_{t+\Delta t} - W_t)/\Delta t$ has standard deviation $1/\sqrt{\Delta t} \rightarrow \infty$ as $\Delta t \rightarrow 0$.

3 Itô integration

We construct $\int_0^T H_t dW_t$ for processes H_t adapted to the Brownian filtration $\{\mathcal{F}_t\}$ (the information generated by W up to time t).

Definition 10 (Filtration and adaptedness) A **filtration** $\{\mathcal{F}_t\}_{t \geq 0}$ is an increasing family of σ -algebras, $\mathcal{F}_s \subseteq \mathcal{F}_t$ for $s \leq t$, modelling information available by time t . A process H_t is **adapted** if H_t is $\mathcal{F}_t$ -measurable for every t — its value at time t depends only on information available by time t .

Definition 11 (Itô integral of a simple integrand) Let H be a **simple** adapted process, constant on a partition $0 = t_0 < t_1 < \dots < t_n = T$ with value $H_{t_{k-1}}$ on $(t_{k-1}, t_k]$. Its Itô integral is

$$\int_0^T H_t dW_t = \sum_{k=1}^n H_{t_{k-1}} (W_{t_k} - W_{t_{k-1}}).$$

The integrand is evaluated at the **left** endpoint of each subinterval. The general Itô integral is the mean-square limit of such sums as the partition is refined.

Theorem 12 (Itô isometry) For an adapted process H with $\mathbb{E}\left[\int_0^T H_s^2 ds\right] < \infty$,

$$\mathbb{E}\left[\left(\int_0^T H_s dW_s\right)^2\right] = \mathbb{E}\left[\int_0^T H_s^2 ds\right].$$

Proposition 13 (Zero-expectation property) Under the same conditions,

$$\mathbb{E}\left[\int_0^T H_s dW_s\right] = 0.$$

Indeed, for a simple integrand each term has $\mathbb{E}[H_{t_{k-1}}(W_{t_k} - W_{t_{k-1}})] = \mathbb{E}[H_{t_{k-1}}] \mathbb{E}[W_{t_k} - W_{t_{k-1}}] = 0$, using independence of the left-endpoint value from the future increment.

4 Itô's formula

Theorem 14 (Itô's formula) Let $f \in C^2$ and let X_t satisfy $dX_t = \mu_t dt + \sigma_t dW_t$. Then

$$df(X_t) = f'(X_t) dX_t + \frac{1}{2} f''(X_t) \sigma_t^2 dt.$$

Expanded into drift and diffusion,

$$df(X_t) = \left(f'(X_t)\mu_t + \frac{1}{2}f''(X_t)\sigma_t^2\right) dt + f'(X_t)\sigma_t dW_t.$$

Remark The extra term $\frac{1}{2}f''(X_t)\sigma_t^2 dt$ — the **Itô correction** — does not appear in classical calculus. It arises from $(dW)^2 = dt$: the second-order term in the Taylor expansion of f is not negligible, because the quadratic variation of W is of order dt , not $(dt)^2$.

Definition 15 (Drift and diffusion) For $dY_t = a_t dt + b_t dW_t$, the coefficient a_t is the **drift** (the dt term) and b_t is the **diffusion** (the dW term). Taking expectations kills the diffusion term (zero-expectation property), so

$$\frac{d}{dt}\mathbb{E}[Y_t] = \mathbb{E}[a_t].$$

This computes expectations of transformed processes: apply Itô's formula, then take expectations to remove the dW term.

5 Stochastic differential equations

Definition 16 (Stochastic differential equation) An **SDE** is

$$dX_t = \mu(t, X_t) dt + \sigma(t, X_t) dW_t, \quad X_0 \text{ given,}$$

with **drift coefficient** μ and **diffusion coefficient** σ . A solution is a process X_t for which $X_t = X_0 + \int_0^t \mu(s, X_s) ds + \int_0^t \sigma(s, X_s) dW_s$.

Theorem 17 (Existence and uniqueness) If μ and σ satisfy the **Lipschitz condition**

$$|\mu(t, x) - \mu(t, y)| + |\sigma(t, x) - \sigma(t, y)| \leq K|x - y|$$

and the **linear growth condition**

$$|\mu(t, x)| + |\sigma(t, x)| \leq K(1 + |x|)$$

for a constant K , then the SDE has a unique solution.

Proposition 18 (Geometric Brownian motion) The SDE

$$dS_t = \mu S_t dt + \sigma S_t dW_t$$

has the explicit solution, via the substitution $f(S) = \ln S$ (so $f'(S) = 1/S$, $f''(S) = -1/S^2$),

$$S_t = S_0 \exp\left(\left(\mu - \frac{1}{2}\sigma^2\right)t + \sigma W_t\right).$$

Itô's formula gives $d(\ln S_t) = (\mu - \frac{1}{2}\sigma^2) dt + \sigma dW_t$; the correction term $-\frac{1}{2}\sigma^2 dt$ is the Itô term.

Proposition 19 (Ornstein-Uhlenbeck process) The SDE

$$dX_t = \theta(m - X_t) dt + \sigma dW_t, \quad \theta > 0,$$

has the solution

$$X_t = m + (X_0 - m)e^{-\theta t} + \sigma \int_0^t e^{-\theta(t-s)} dW_s.$$

The drift $\theta(m - X_t)$ pulls X_t toward the level m at rate θ (**mean reversion**). As $t \rightarrow \infty$ the **stationary distribution** is

$$\mathcal{N}\left(m, \frac{\sigma^2}{2\theta}\right).$$

Definition 20 (CIR model) The **Cox-Ingersoll-Ross** SDE adds a square-root diffusion to mean reversion:

$$dX_t = \theta(m - X_t) dt + \sigma\sqrt{X_t} dW_t.$$

The factor $\sqrt{X_t}$ makes the diffusion vanish as $X_t \rightarrow 0$, keeping the process nonnegative.

6 The Black-Scholes model

The stock price follows geometric Brownian motion $dS_t = \mu S_t dt + \sigma S_t dW_t$. A derivative has price $V(t, S_t)$.

Theorem 21 (Black-Scholes PDE) Construct a portfolio long one derivative and short $\partial V / \partial S$ shares (**delta hedging**). Applying Itô's formula to $V(t, S_t)$ and choosing the hedge to cancel the dW term, the portfolio is riskless and must earn the risk-free rate r . This yields

$$\frac{\partial V}{\partial t} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + r S \frac{\partial V}{\partial S} - rV = 0.$$

Definition 22 (Risk-neutral valuation) Under the **risk-neutral measure** Q , the drift of S is replaced by the risk-free rate, $\mu \rightarrow r$:

$$dS_t = r S_t dt + \sigma S_t dW_t^Q.$$

The price of a derivative with payoff $\Phi(S_T)$ is the discounted expected payoff

$$V(t, S_t) = e^{-r(T-t)} \mathbb{E}^Q[\Phi(S_T) \mid \mathcal{F}_t].$$

Theorem 23 (Black-Scholes formula) For a European call with strike K and maturity T , write

$$d_1 = \frac{\ln(S/K) + (r + \frac{1}{2}\sigma^2)(T-t)}{\sigma\sqrt{T-t}}, \quad d_2 = d_1 - \sigma\sqrt{T-t}.$$

Then, with N the standard normal CDF,

$$C = S N(d_1) - K e^{-r(T-t)} N(d_2).$$

Definition 24 (Greeks) The **Greeks** are sensitivities of the option price. For the European call:

- **Delta:** $\partial C / \partial S = N(d_1)$.
- **Gamma:** $\partial^2 C / \partial S^2 = N'(d_1) / (S \sigma \sqrt{T-t})$.
- **Vega:** $\partial C / \partial \sigma = S N'(d_1) \sqrt{T-t}$.

Proposition 25 (Put-call parity) For European call C and put P with the same strike K and maturity T ,

$$C - P = S - K e^{-r(T-t)}.$$

7 Monte Carlo option pricing

Risk-neutral valuation expresses a price as an expectation $\mathbb{E}^Q[e^{-rT} \Phi(S_T)]$, which is estimated by simulation.

Definition 26 (Euler-Maruyama discretization) For $dX_t = \mu(t, X_t) dt + \sigma(t, X_t) dW_t$, fix a step Δt and simulate

$$X_{t+\Delta t} = X_t + \mu(t, X_t)\Delta t + \sigma(t, X_t)\sqrt{\Delta t} Z, \quad Z \sim \mathcal{N}(0, 1).$$

The diffusion term scales as $\sqrt{\Delta t}$ (not Δt), reflecting the Brownian increment $W_{t+\Delta t} - W_t \sim \mathcal{N}(0, \Delta t)$.

Definition 27 (Monte Carlo estimator) Simulate N independent terminal values $S_T^{(1)}, \dots, S_T^{(N)}$ under Q and average the discounted payoff:

$$\hat{V} = \frac{1}{N} \sum_{i=1}^N e^{-rT} \Phi(S_T^{(i)}).$$

By the law of large numbers $\hat{V} \rightarrow V$, and by the CLT a **confidence interval** is

$$\hat{V} \pm z_{\alpha/2} \frac{s}{\sqrt{N}},$$

with s the sample standard deviation of the discounted payoffs.

Definition 28 (Antithetic variates) For each simulated path using normals Z , also use $-Z$ to generate a second path, and average the two payoffs. The negative correlation between the paired estimates reduces the variance of the average.

Definition 29 (Control variates) Let Y be the quantity to estimate and X a correlated quantity with known mean $\mathbb{E}[X]$. Use

$$Y_c = Y - c(X - \mathbb{E}[X]),$$

which has the same mean as Y ; choosing $c = \text{Cov}(Y, X) / \text{Var}(X)$ minimizes $\text{Var}(Y_c)$, reducing the estimator's variance.